INTELLIGENT CAR POOLING SYSTEM

PROJECT REPORT

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to

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of
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Department of Computer Science and Engineering
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Kerala

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DECLARATION

We undersigned hereby declare that the project report **Intelligent Car Pooling System**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of **Prof. Akhila Vijay**. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Place: Thiruvananthapuram
Date: April 1, 2026

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CERTIFICATE

This is to certify that the report entitled **"INTELLIGENT CAR POOLING SYSTEM",** submitted by **E S Sahal Hussain** (TVE23CS054), **Hani Muhamed Ashraf** (TVE23CS067), **Johan Antony Davis** (TVE23CS075) to the **APJ Abdul Kalam Technological University** in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Prof. Akhila Vijay	**Prof. Ramcy R**	**Dr. Ajeesh Ramanujan**
Assistant Professor	Assistant Professor	Associate Professor
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(Guide)	(Coordinator)	(Head of Department)

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E S Sahal Hussain
Hani Muhamed Ashraf
Johan Antony Davis

ABSTRACT

Rising fuel costs, traffic congestion, and underutilised vehicles make solo commuting increasingly unsustainable. While carpooling presents a viable alternative, it is widely underused due to the lack of accessible coordination tools, real-time matching capabilities, and inherent trust barriers. To address these issues, this project introduces an Intelligent Car Pooling System—a Peer-to-Peer web application developed using the MERN (MongoDB, Express.js, React.js, Node.js) stack.

The system is designed to provide clear route, schedule, and availability information to encourage carpool usage among daily commuters, university students, and staff. Key features include a 3-stage matching algorithm (bounding box, seat check, and Mapbox Matrix detour feasibility), interactive route visualisation using Mapbox GL JS, and a secure dual JWT-based authentication system. By effectively bridging the information gap and automating the matching process, the proposed application aims to optimise route overlaps, reduce travel costs, and decrease overall carbon emissions.

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ABBREVIATIONS

(List in the alphabetical order)

API Application Programming Interface

MERN MongoDB, Express.js, React.js, Node.js

UI User Interface

Chapter 1
Introduction

Rising fuel costs and traffic congestion make solo commuting increasingly unsustainable in modern urban environments. While carpooling presents a practical solution to reduce the number of vehicles on the road, it remains widely underused. This is primarily due to the lack of accessible coordination tools and real-time ride-matching capabilities. Underutilised vehicles directly contribute to increased road congestion and higher carbon emissions. This project targets daily commuters, university students, and staff members, aiming to provide a seamless digital platform for ride-sharing.

1.1 Exact Problem Context

Manual coordination of carpooling through calls, messages, or physical noticeboards is unreliable, time-consuming, and fails to scale effectively. There exists a significant information gap where users cannot easily discover rides with similar routes or schedules. Furthermore, the absence of real-time matching prevents the accurate calculation of route overlaps or detour costs.

A lack of structured verification layers also creates a trust barrier, making riders hesitant to share rides with strangers. Because trip timings are not centralised, commuters experience missed matches and often abandon carpooling entirely in favour of individual vehicles.

MongoDB, Express.js, React.js, Node.js (MERN), Application Programming Interface (API) and User Interface (UI) are abbreviations used throughout this report.

Chapter 2
Existing Methods

This chapter explores various existing methodologies and studies related to carpooling systems to understand the current landscape, identify technological barriers, and establish areas for improvement.

2.1 Algorithmic Optimization Methods

Several studies have focused heavily on algorithmic routing. Beed et al. (2020) proposed a Hybrid Multi-Objective Carpool Route Optimization technique using a Genetic Algorithm and A* Algorithm. While this approach minimizes detour and pickup costs effectively and outperforms standard Genetic Algorithms, it involves a highly complex setup that is not well-suited for real-time web applications.

2.2 Institutional and Social Approaches

Colorni & Luè (2011) introduced PoliUniPool, a system designed specifically for universities featuring time-window constraints and cost-sharing. However, it suffers from a restricted user base and lacks a dedicated mobile interface. Similarly, Xia et al. (2019) developed a Social and Route Network Model. While this model improved trust by ten times and reduced social obstructions, it resulted in a 6.3% route cost increase and heavily relies on acquiring pre-existing social graph data.

Other models, such as the network-based matching by Xia et al. (2015), handle multiple carpools but face high combinatorial complexity, making them slow for large networks.

2.3 Motivation

The study by Adélè Dionisio (2020) found that overly complex "smart" features in carpooling apps can confuse users. This highlights the need for a simple, responsive, real-time peer-to-peer carpooling web application that overcomes trust and information barriers without overwhelming users.

Chapter 3
Problem Statement and Objectives

To overcome the friction points identified in existing literature, this project proposes a modernized approach to ride-sharing coordination.

3.1 Problem Statement

The problem is stated as to design and develop a web application that provides clear carpooling route, schedule, and availability information to encourage its usage. Current manual and fragmented scheduling systems are insufficient, requiring a centralized, intelligent digital solution.

3.2 Objectives

The primary objectives of this project are:

- To develop a secure Peer-to-Peer Carpooling Web Application using the MERN stack.
- To implement a 3-stage matching algorithm (bounding box, seat check, detour feasibility) for discovering compatible rides.
- To provide interactive, real-time route visualization using Mapbox GL JS.
- To create a secure authentication system using dual JWT tokens to build user trust.
- To enable drivers to seamlessly post trips, manage requests, and invite co-passengers.

Chapter 4
Design and Implementation

This chapter outlines the system architecture, modules, and the specific technology stack utilized to build the Intelligent Car Pooling System.

4.1 System Architecture

The system follows a modern client-server architecture utilizing the MERN stack. The frontend is built with React.js and Tailwind CSS for a highly responsive user interface. Mapbox GL JS SDK and Directions API are integrated for mapping and route visualization. The backend is powered by Node.js and Express.js, with MongoDB serving as the NoSQL database for flexible document storage.

4.2 Module Description

The application consists of several core functional modules:

- **Auth Module:** Handles user registration and login using bcrypt for password hashing and dual JWT tokens (1-hour access, 1-day refresh) for stateless authentication.
- **Trip Posting Module:** Validates origin, destination, date, seats, and maximum detour inputs. It fetches the base duration via the Mapbox Directions API before saving the confirmed listing to MongoDB.
- **Matching Engine:** Runs a 3-stage filter utilizing the Mapbox Matrix API ($N \times N$ matrix) to compute permutations and determine optimal stop ordering and detour feasibility.
- **Map Visualisation Module:** Renders a blue driver route and a green rider pickup-dropoff overlay using Mapbox GL JS.
- **Booking & Driver Management:** Manages incoming seat requests, recalculates route durations upon driver acceptance, and updates the trip document

accordingly.

4.3 System Diagrams

To provide a comprehensive visual understanding of the system's design and data lifecycle, the architectural diagram and corresponding Data Flow Diagrams (DFDs) are presented below.

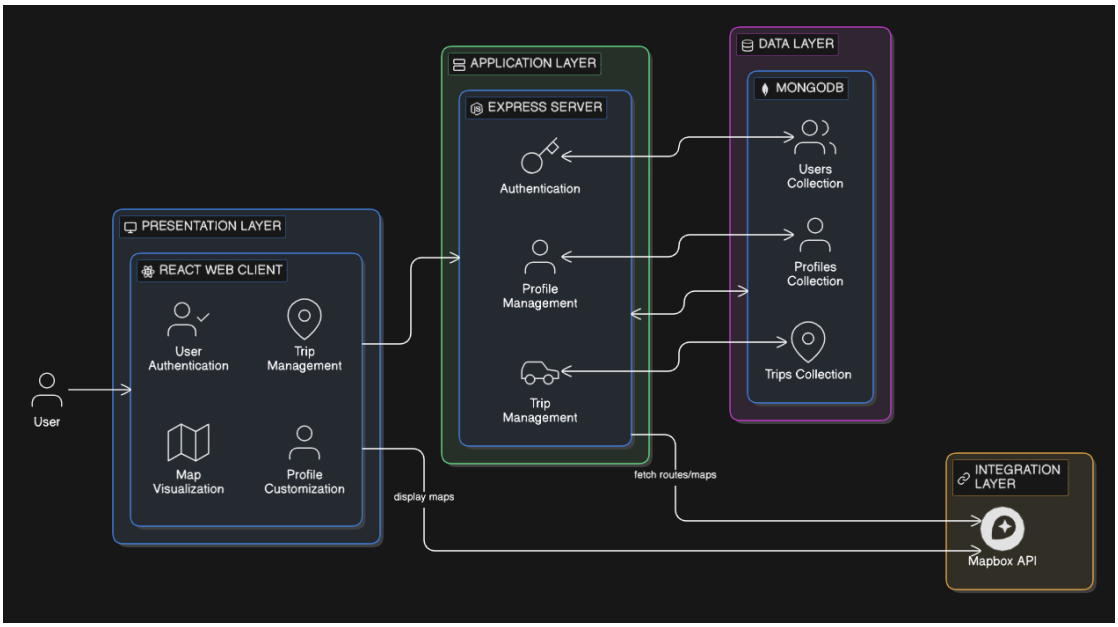

Figure 4.1: System Architecture Diagram

4.3.1 Data Flow Diagrams

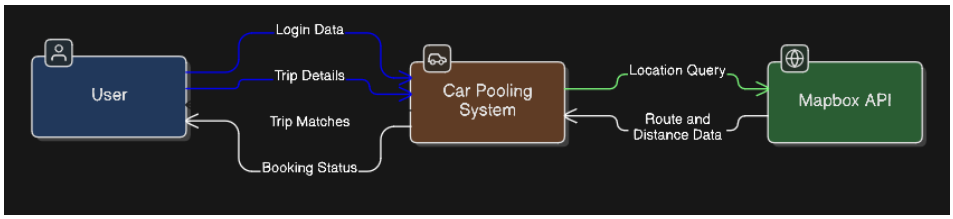

Figure 4.2: Level 0 Data Flow Diagram (Context Diagram)

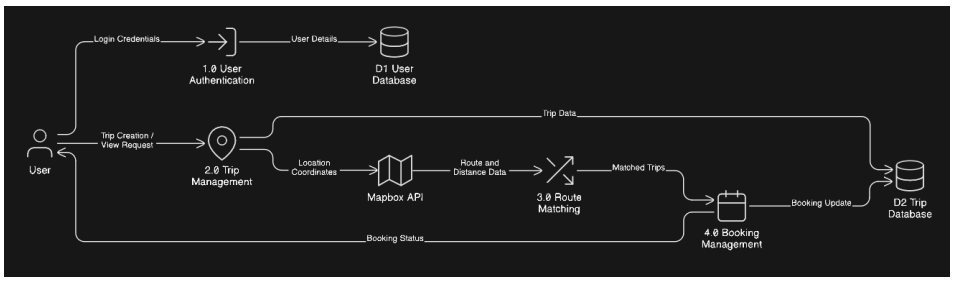

Figure 4.3: Level 1 Data Flow Diagram

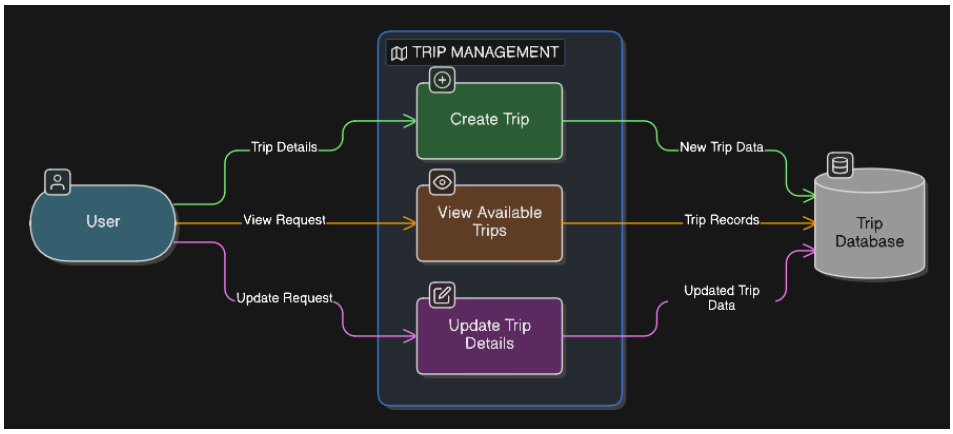

Figure 4.4: Level 2 Data Flow Diagram

Chapter 5
Results and Discussion

The Intelligent Car Pooling System has been successfully developed, integrated, and tested according to the planned work schedule.

5.1 Module Integration and Visualization

All core functional modules—including Authentication, Trip Posting, the Matching Engine, and Booking—are 100% integrated. The system successfully renders Mapbox GL maps with dynamic route polylines. The interactive map accurately differentiates layered routes, providing immediate visual feedback to both drivers and riders.

System Module	Completion Status	Key Features Achieved
Auth & Profile	100%	Bcrypt hashing, Dual JWT tokens, Image Resizing
Trip Management	100%	Mapbox Directions API, MongoDB CRUD
Matching Engine	100%	3-stage filter, Matrix API routing
User Interface	100%	Fully responsive Tailwind CSS layout

Table 5.1: Project Module Completion and Integration Status

5.2 Performance and Data Consistency

MongoDB Atlas successfully handles all read and write operations efficiently, maintaining data consistency across users and trips. The stateless authentication system ensures secure sessions without server-side memory overhead, contributing to a lightweight and fast application backend.

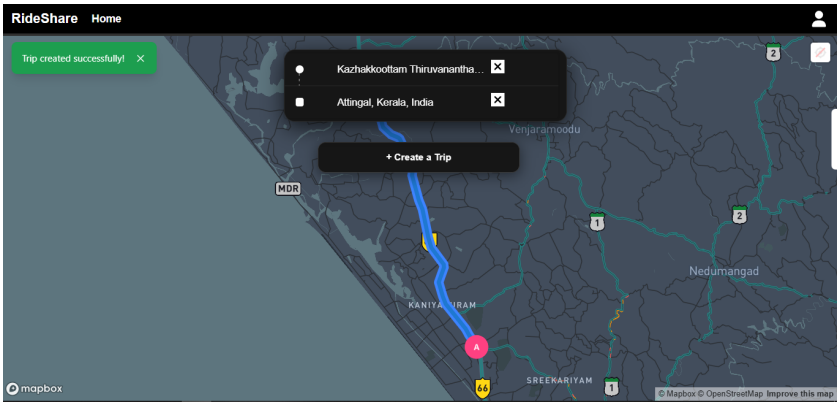

Figure 5.1: User A creating trip

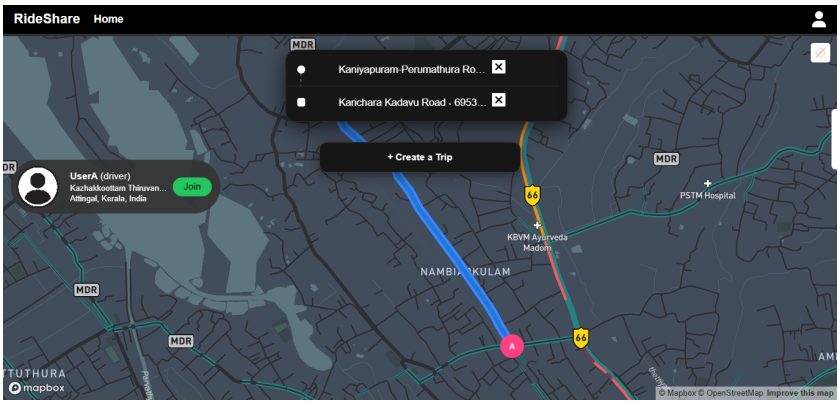

Figure 5.2: User B has option to request carpool to User A

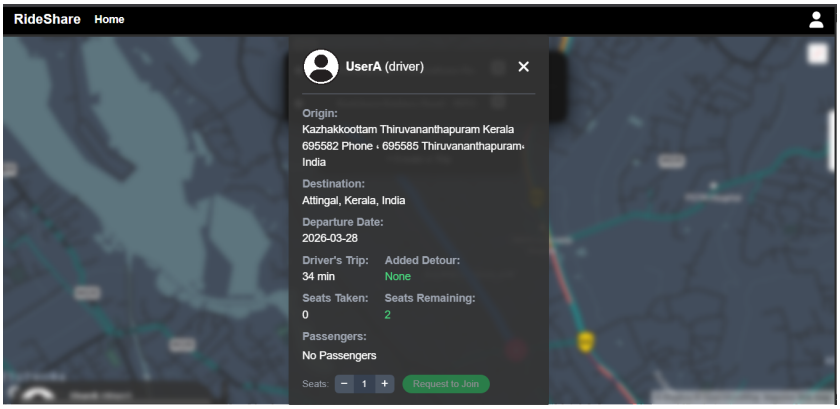

Figure 5.3: User B request for carpool

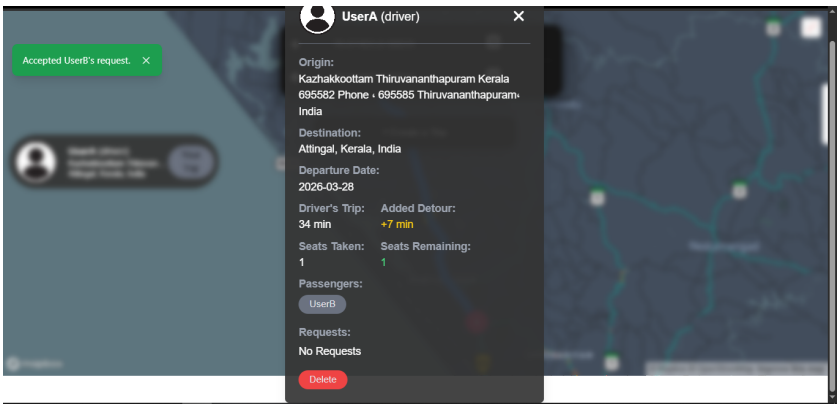

Figure 5.4: User A accept request of User B

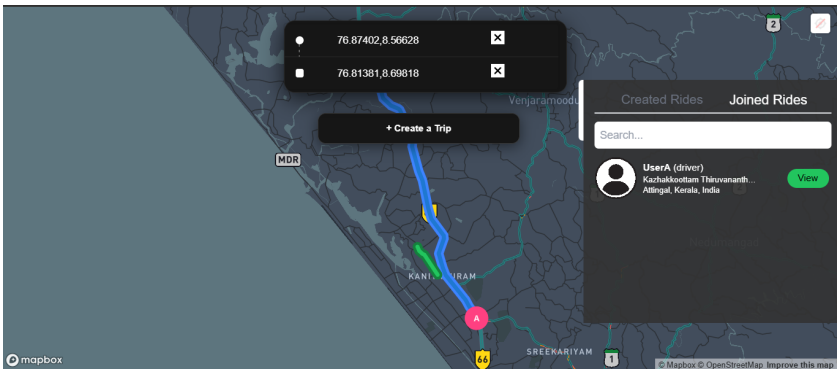

Figure 5.5: User B visualize the two routes

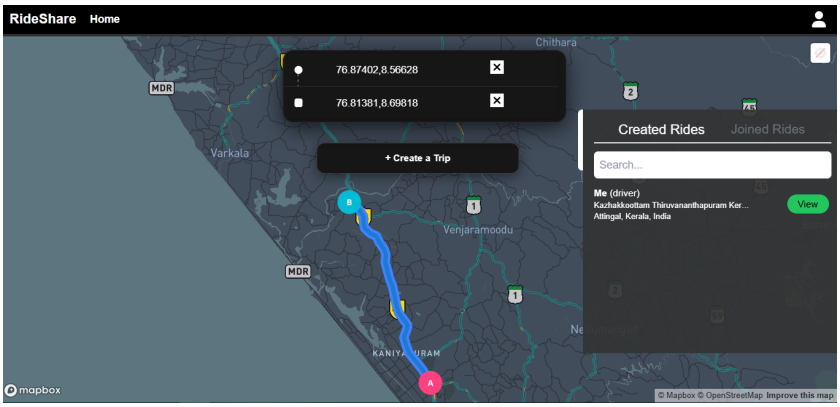

Figure 5.6: New route for User A

Chapter 6
Conclusion and Future Scope

6.1 Conclusion

The Intelligent Car Pooling System successfully addresses the challenges of fragmented scheduling and lack of real-time matching in daily commutes. By utilizing the MERN stack architecture, the project provides a seamless digital solution. The dual JWT authentication system ensures security, while the 3-stage matching engine optimizes route overlaps and validates detour feasibility. The successful implementation of interactive mapping via Mapbox GL JS demonstrates the viability of a robust, peer-to-peer carpooling platform designed to reduce road congestion.

6.2 Future Scope

While the present system meets its primary objectives, there are several areas targeted for future enhancement:

- **Scalability Optimization:** Addressing the factorial growth in Mapbox Matrix API permutation counts as the number of riders in a single pool increases.
- **Real-time Updates:** Transitioning from standard polling to WebSocket-based connections for instantaneous user notifications regarding ride requests and approvals.
- **Trust and Safety Enhancements:** Implementing a comprehensive user rating system and official identity verification to further lower the trust barrier.
- **Mobile Responsiveness:** Further optimizing Tailwind CSS layouts to ensure flawless display and interaction across highly varied and smaller mobile screen sizes.

References

[1] R. S. Beed, S. Sarkar, A. Basu, and D. Roy, "A Hybrid Multi-Objective Carpool Route Optimization Technique using Genetic Algorithm and A* Algorithm," *arXiv preprint arXiv:2009.04365*, Sep. 2020.

[2] A. Colorni and A. Luè, "PoliUniPool: A Carpooling System for Universities," *Proc. Social Behav. Sci.*, vol. 20, pp. 558–567, 2011.

[3] S. Adélé and C. Dionisio, "Learning from the Real Practices of Users of a Smart Carpooling App," *Eur. Transp. Res. Rev.*, vol. 12, no. 1, pp. 1–13, 2020.

[4] J. Xia, K. M. Curtin, D. Wu, and W. Li, "A Carpool Matching Model with Both Social and Route Networks," *Transp. Res. C, Emerg. Technol.*, vol. 104, 2019.

[5] J. Xia, K. M. Curtin, W. Li, and Y. Zhao, "A New Model for a Carpool Matching Service," *PLOS ONE*, vol. 10, no. 6, p. e0129257, Jun. 2015.

[6] J. V. Joshy, N. Kanakarajan, N. M. John et al., "Ride Sharing," *Int. J. Eng. Res. Technol.*, vol. 12, 2023.

[7] J.-B. Lee, "Company-Wide Carpooling for Long Distance Commuting in South Korea and Its Effects on Reducing Transportation Problems," *Sustainability*, Springer, 2022.

[8] K. A. Soltys, "Toward an Understanding of Carpool Formation and Use," M.S. thesis, Dept. Geography, Univ. of Toronto, Toronto, ON, Canada, 2009.

[9] G. N. Q. Carvalho, J. R. K. Procópio et al., "Urban Mobility of Medium-Sized Cities: A Case Study on the Carpool System at Unifei, Itabira Campus," *RSD J.*, 2020.

[10] K. Sheoran, V. Jatana, R. Gulati et al., "Intelligent Transportation System Architecture for a Carpool System," *Int. J. Comput. Appl.*, 2014.

[11] Q. Xiao and R. He, "Carpooling Scheme Selection for Taxi Carpooling Passengers: A Multi-Objective Model and Optimisation Algorithm," *Arch. Transp.*, 2017.

[12] A. Aguiéra and E. Pigalle, “The Future and Sustainability of Carpooling Practices: An Identification of Research Challenges,” *Sustainability*, vol. 13, no. 19, p. 10817, MDPI, Sep. 2021.

[13] K. Jaybhaye, A. Chavan et al., “A Literature Survey on Carpooling,” *Int. J. Adv. Res. Sci. Commun. Technol.*, 2023.

[14] “Car-Pooling System (CapX),” *Int. J. Adv. Res. Sci. Commun. Technol.*, 2022.

[15] M. E. S. Kamel, “A Requirement Engineering Approach to a Carpool System for University Mobility,” *OASK Publ.*, 2025.

[16] “Carpooling Adoption for Educational Trips toward Sustainable Mobility,” in *Proc. ACM Conf.*, 2022.

[17] “Time-Optimal and Privacy-Preserving Route Planning for Carpool Policy,” *Springer*, 2022.

[18] “What Encourages People to Carpool? A Conceptual Framework of Carpooling Psychological Factors and Research Propositions,” *Transp. Res. Interdiscip. Perspect.*, 2021.

[19] R. Teal, “Carpooling: Who, How and Why,” *Transp. Res. A, Gen.*, 1987.

[20] T. Siyuan, D. Renjie et al., “Commute with Community: Enhancing Shared Travel through Social Networks,” *arXiv preprint*, 2024.